# Structural Imperceptibility and the Failure of Climate Action

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## Introduction

We have known about the greenhouse effect for well over a century. Svante Arrhenius calculated as early as 1896 that doubling atmospheric carbon dioxide would raise global temperatures by several degrees.[1] Since then we have not only learned more but emitted more. Scientists have reached an unequivocal consensus that human activity is the dominant cause of observed warming since the mid-twentieth century.[2] However, this increase in information over the years did not bring a response proportionate to the problem. We have grown from 280 ppm preindustrially to over 430 currently.[3] There have been a few moments of climate policy and action lurching forward, but currently we are marred with more degenerate measures. The usual explanation for our lack of action or progress is that people need more information, better communication, or more vivid images of catastrophe.

The truth is in that, but it does not go far enough. Climate action consistently falls short not because people do not care, but because climate change is unusually hard to apprehend at the scale of ordinary human experience. Carbon dioxide is invisible, its harms are diffuse, and its more serious effects are delayed long enough to detach cause from consequence in everyday moral life. The problem does not announce itself the way smoke, poisoned water, or a burning river does. It helps explain not only the paralysis, but misdirection. When climate concern does register, it often gets channeled toward the most visible and controllable thing in sight such as individual behavior. The result is a moral culture organized around personal carbon footprints, consumer choices, and lifestyle adjustments, even though the overwhelming share of emissions flows through industrial systems of extraction and production that no private consumer can directly govern. This paper argues that climate action consistently falls short not because people lack concern but because climate change is structurally imperceptible at human scale—invisible, diffuse, and temporally distant—which both paralyzes motivation and misdirects moral energy toward individual behavior while the overwhelming share of emissions flows from industry; effective response therefore requires political leadership that makes the crisis legible and an honest reckoning with where emissions actually come from. The focus here is on political leaders not because markets, religious institutions, or civil society are irrelevant, but because political leaders possess the institutional authority to redirect collective pressure at the level where binding change can occur. They can restrict extraction, reorganize incentives, create durable institutions, and force future costs into present decision-making. A further question remains about practical substitutes and viable energy transitions—which surely matter—but it lies

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[1] Svante Arrhenius, "On the Influence of Carbonic Acid in the Air upon the Temperature of the Ground," *Philosophical Magazine and Journal of Science* 41, no. 251 (1896): 237–76.
[2] IPCC, "Summary for Policymakers," in *Climate Change 2021: The Physical Science Basis*, ed. V. Masson-Delmotte et al. (Cambridge: Cambridge University Press, 2021), SPM-5.
[3] NOAA Global Monitoring Laboratory, "Trends in Atmospheric Carbon Dioxide," accessed May 2026, https://gml.noaa.gov/ccgg/trends/.

beyond this paper's scope. Workable substitutes will remain politically underused unless the crisis is made legible enough to compel structural action.

## Why Climate Change is Structurally Imperceptible

The failure of climate action is often described as a failure of education. People do not know enough, trust science enough, or visualize the stakes clearly enough. That explanation misses the makeup of the problem itself. Climate change is difficult to respond to not simply because it is complicated, but because its central causal features evade the perceptual and emotional systems through which human beings usually recognize danger and organize response. There are three features here that matter most. First, carbon dioxide is invisible and odorless. A person cannot look at the sky and see greenhouse gases accumulating there. Second, its harms are diffuse. A given emission does not map cleanly onto a single victim, place, or moment. The atmosphere mixes globally, so the causal trail from act to injury is dispersed beyond ordinary experience. Third, climate change is temporally stretched. As Deane Curtin emphasizes, the effects of emissions unfold with such delay that present action and future consequence come apart in lived experience; one can help create a harm one will never personally witness in full.[4] There is roughly a forty-year lag between emission and its full thermal effect, which means we are already experiencing the consequences of decisions made in the 1980s and generating consequences for people who have not yet been born. That kind of lag does not work well for moral motivation and leads to controversial discounting conversations.

The Montreal Protocol is a useful contrast because it shows what environmental politics looks like when those conditions are more favorable. Ozone depletion could be rendered legible in a way climate change resists. It had a neat, visible image of the ozone hole. It was linked to familiar harms like skin cancer and ultraviolet exposure. Its principal industrial drivers were comparatively narrow, and substitutes for chlorofluorocarbons were available quickly enough to support coordinated policy.[5] This does not mean the perceptibility alone explains the Protocol's success; the availability of substitutes and narrower industrial targets also mattered. But those conditions became politically usable because the underlying problem was legible in the first place. Climate change offers none of that with how extensive it is. Robin Globus Veldman's fieldwork in Georgia sharpens the point. In the communities she studied, concern about mercury contamination in the local waterways was real, practical, and active. Concern about atmospheric carbon was not. She summarizes, church members who were "unconcerned about the global problem of carbon dioxide in the air" were nonetheless deeply concerned about "the local issue of mercury in the water" because it threatened fish they actually caught and ate.[6] This is a

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[4] Deane Curtin, "To Live as a Lotus among the Flames: Buddhist Awakening in the Middle of the Climate Crisis," *Worldviews* 21 (2017): 23.

[5] *Montreal Protocol on Substances that Deplete the Ozone Layer, September 16, 1987, United Nations Treaty Series 1522: 3.*

[6] Robin Globus Veldman, *The Gospel of Climate Skepticism: Why Evangelical Christians and the Environment Don't Mix* (Oakland: University of California Press, 2019), 57.

clarifying comparison because it strips away the explanation that the difference must be ignorance or indifference. The same people could respond to one environmental threat and ignore another because the threats appeared at different scales of perception. Mercury threatened bodies, meals, and local places; carbon only threatened a planetary system on a scale no one can directly see.

The deeper implication is that climate change exceeds not only common sense but inherited habits of moral attention. Mark Hathaway, drawing on the neurologist Antonio Damasio, describes a kind of collective "myopia for the future" in which societies discount delayed harms in favor of immediate comfort and continuity.[7] Curtin puts it more bluntly via Dale Jamieson, evolution did not design human beings to recognize or solve a crisis of this kind. We are tuned to movement, immediacy, and visible disruption. "We have a strong bias toward dramatic movements of middle-sized objects that can be visually perceived, and climate change does not typically present in this way."[8] From an evolutionary standpoint, the greatest threats our ancestors faced were immediate, and the perceptual systems that developed in response are calibrated for that environment. Greenhouse gas accumulation falls entirely outside it. The emotional consequence of that mismatch makes the political problem worse. Hathaway argues that large-scale ecological threats often trigger not sustained engagement, but paralysis, denial, and psychic numbing.[9] Fear does not automatically mobilize. When the threat is too large, too delayed, or too difficult to translate into effective agency, fear can become a reason to look away. That aversion helps explain why climate communication so often oscillates between two failures: technical explanation that never grips people at all, and catastrophic imagery that grips them briefly and then leaves them overwhelmed. Better graphs do not solve a scale mismatch. More terrifying projections do not automatically create durable action. Climate change is difficult because it asks human beings and political communities to respond to a danger that does not present itself in the sensory and temporal form to which they are best adapted. Any serious climate ethics must begin there.

## How Misperception Misdirects Moral Energy

Structural imperceptibility does not only weaken motivation, but it also bends it toward the wrong object. With climate change being so hard to see directly, the moral energy it generates looks for visible point of contact. Most often it lands on the individual self—what I drive, what I eat, if I recycle, how often I fly, is my consumption responsible. The problem with this framework is not that personal behavior is irrelevant, it is that it mistakes the most available site of moral action for the most consequential one. The empirical balance is hard to ignore. In 2017, Paul Griffin's Carbon Majors analysis found that 71% of industrial greenhouse gas emissions

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[7] Mark D. Hathaway, "Overcoming Fear, Denial, Myopia, and Paralysis: Scientific and Spiritual Insights into the Emotional Factors Affecting Our Response to the Ecological Crisis," *Worldviews* 21 (2017): 176.
[8] Curtin, "Lotus among the Flames," 23, drawing on Dale Jamieson, *Reason in a Dark Time* (Oxford: Oxford University Press, 2014).
[9] Hathaway, "Overcoming Fear," 176–78.

since 1988 can be traced to one hundred fossil fuel producers, including the downstream combustion of the fuels they extracted and sold.[10] Of course, individuals are not metaphysically disconnected from emissions, it is that the bulk of the carbon economy is organized through concentrated systems of production, licensing, extraction, transport, and investment. Michael Northcott makes the political implication plain, focusing on how household restraint obscured the fact that fossil fuel extraction is controlled by corporations and states, and reduced consumption by some users does not by itself keep reserves in the ground.[11] Add Jevons Paradox, in which efficiency gains are often absorbed by increased total consumption, and the limits of consumer-centered ethic become harder to deny.[12]

Willis Jenkins helps explain why this misdirection has proven so durable. Climate change, he argues, exposes the inadequacy of inherited ethical frameworks for harms that are indirect, cumulative, and distributed across generations and institutions.[13] Those frameworks were built for a different kind of moral situation—one in which a harm can be traced to an identifiable agent acting with some relevant intention toward an identifiable victim. On that model, moral responsibility tracks direct causation. Climate change defeats it at every step. The commuter who drives to work, the tenant who heats an apartment, or the parent who buys groceries rarely intends climatic harm, cannot isolate their own contribution, and often lacks meaningful alternatives within existing infrastructure. Each of these inherited frameworks captures something real about moral life, but none was built for the kind of problem climate change is. What comes into view instead is responsibility without direct causation. Jenkins turns to Iris Marion Young's social connection model because it is better suited to injustices produced by normal participation in unjust structures.[14] On this account, responsibility does not depend on proving that a particular agent directly caused a particular harm. It arises from participation in social processes that predictably generate unjust outcomes. That shift matters greatly for climate ethics because the central stakes are intergenerational. Future generations have no vote in present institutions; they cannot consent to the risks being imposed on them. Present actors therefore inherit both the damage and decision. Jenkins' point is sharp, the strongest ethical pressure in climate change falls precisely where traditional models of liability are weakest. We owe something to people we did not injure face-to-face and will never meet, and we owe it to them because we are sustaining a system whose harms arrive late, cumulatively, and unequally. Carbon dioxide emissions become morally charged, Jenkins argues, not because turning on a light or driving to work is intrinsically wrongful, but because those ordinary acts now occur within an atmosphere already loaded by past generations and within infrastructures that make

[10] Paul Griffin, *The Carbon Majors Database: CDP Carbon Majors Report 2017* (London: CDP UK, 2017), 8.
[11] Michael S. Northcott, *A Political Theology of Climate Change* (Grand Rapids: William B. Eerdmans, 2013), 202–3.
[12] Northcott, *Political Theology*, 203–4.
[13] Willis Jenkins, *The Future of Ethics: Sustainability, Social Justice, and Religious Creativity* (Washington, DC: Georgetown University Press, 2013), 17, 30–34.
[14] Jenkins, *Future of Ethics*, 34; Iris Marion Young, *Responsibility for Justice* (Oxford: Oxford University Press, 2011).

fossil participation hard to avoid.[15] That complicates both blame and innocence, present people did not single-handedly build the system, but they do reproduce it. Future people did not consent to its risks, but they will absorb them. Climate ethics therefore cannot be satisfied by identifying a villainous intention and condemning it. The more troubling reality is that normal life itself has been organized in ways that distribute responsibility widely while concentrating decisive power elsewhere.

Individual action is not worthless, but its value changes considering that analysis. Personal restraint and changed habits become serious when they connect the self to structural change rather than substituting for it. Individual conduct can matter as witness, solidarity, or movement-building. It matters when it helps create pressure for altered infrastructure, altered law, and altered production. Detached from that larger horizon, lifestyle ethics easily becomes a moral sinkhole, absorbing anxiety that ought to be directed outward. The history of the carbon footprint discourse shows how convenient that sinkhole has been. As Mark Kaufman recounts, BP helped popularize the personal carbon footprint as a public-facing framework for climate responsibility in the early 2000s.[16] The maneuver fit a wider moral instinct Veldman also identifies, the preference for environmental acts that are private, local, and individual rather than political or collective.[17] That preference is not confined to one religious subculture; it saturates the broader discourse. When guilt is privatized, pressure is privatized too. Consumers are told to purify themselves while producers, regulators, and investors escape a comparable moral spotlight. Climate change is a crisis of structural responsibility, delayed consequence, and obligation to people beyond the range of direct reciprocity.

## **Legibility and Leadership**

If climate change is difficult to perceive and easy to moralize at the wrong scale, then some institution has to perform a mediating function. It must make the crisis politically legible, turning a diffuse, long-horizon danger into something concrete enough to organize conflict, justify coercive policy, and bind present actors to future consequences. The institution best positioned to do that is political leadership. This is not because leaders are morally superior; it is because leadership can perform a public legibility function that private actions cannot perform on their own. American environmental history offers several examples. Theodore Roosevelt used federal power to treat conservation as a national trust rather than a private preference. Franklin Roosevelt's conservation programs made ecological stewardship a matter of public work and public purpose. Richard Nixon signed the National Environmental Policy Act, the Clean Air Act, and the Clean Water Act and created the Environmental Protection Agency not out of personal conviction but because political institutions had translated visible pollution into an urgent national problem. Those episodes matter because they show that leadership has repeatedly

[15] Jenkins, *Future of Ethics*, 34.
[16] Mark Kaufman, "The Carbon Footprint Sham," *Mashable*, July 13, 2020, https://mashable.com/feature/carbon-footprint-pr-campaign-sham.
[17] Veldman, *Gospel of Climate Skepticism*, 53–54.

converted underweighted goods into durable policy commitments. The Montreal Protocol shows the same function in an international key. Once ozone depletion had been rendered mappable, medically tangible, and publicly intelligible, governments could move.[18] Again, substitutes and industrial concentration mattered—and they deserve acknowledgement—but a crisis had to become legible before it could become politically actionable. Leadership did not create the science, but it translated the science into an object of coordination.

Northcott's contribution is to explain why that translation has been so much harder for climate change. Modern liberal governance, he argues, is structurally oriented toward the present and the near-term. Electoral cycles reward attention to immediate concerns and visible constituencies, market governance optimizes for current prices that systematically discount future costs, and the liberal political tradition itself—from Hobbes and Locke onward—has constructed legitimate authority around the protection of present persons and present property rather than future generations or the ecological system on which all persons depend.[19] Northcott traces this to something deeper than institutional design, to the foundational move in modern political thought of constructing political authority independently of nature, treating the non-human world as a resource or backdrop to be extracted rather than a constitutive element of political community. The result is what he calls a "crossed-out" nature—governance built on a premise of human separation from ecological systems, and therefore structurally blind to a crisis defined by the transgressions of those systems' limits. The problem is not that politicians lack good intentions, it is that the institutions through which they act were not built to see the kind of problem climate change actually is. This structural incapacity is compounded by what Northcott calls the "slow catastrophe" problem.[20] Democratic politics is organized around crises, moments of visible, urgent threat that mobilize public attention and political will—eerily parallel to our own evolutionary wiring. Climate change produces no such moment. Its harms accumulate gradually, unevenly, and with enough ambiguity about attribution that no single event triggers the kind of response that something like Pearl Harbor or the 2008 financial crisis triggered. By the time the cumulative effects are dramatic enough to be unambiguously legible, the window for effective responses will have significantly narrowed. Political institutions calibrated to respond to crises are being asked to prevent one that announces itself only in the data.

What would adequate political leadership actually look like in this context? At a minimum, it would require leaders who function as perception-proxies, using the authority and visibility of political office to make present in political life the harms and consequences that structural imperceptibility keeps distant and abstract. This means treating the crisis not as a technical problem to be delegated to agencies and managed through voluntary targets, but as a political fact worth organizing around. It would also require redirecting moral and regulatory pressure away from individual behavior and toward the extraction and production decisions that

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[18] *Montreal Protocol.*
[19] Northcott, *Political Theology*, 204–8.
[20] Northcott, *Political Theology*, 5–6, 8, 12.

drive emissions, through carbon pricing, extraction limits, fossil fuel subsidy elimination, and binding sectoral targets.[21] It would require building institutions capable of deliberating across time horizons longer than electoral cycles, structures which can hold obligations to future generations present in current political decision-making in the way that constitutional courts—are at least supposed to—hold past commitments present across generations. None of this is politically simple. The interests aligned against each of these moves are substantial, and the perceptual problem that makes climate change difficult to motivate in the first place also makes it difficult to sustain political will against well-organized opposition. But diagnosing the structural problem is not the same as pronouncing it insoluble. The Montreal Protocol, the creation of the EPA, and the national park system each required political leadership to do exactly what this analysis calls for, making a public good politically real, redirecting pressure toward the actors responsible, and absorbing the opposition that followed. That those examples were possible under more favorable perceptual conditions does not mean the function itself is unavailable. It means the challenge of performing it is—under the current conditions—considerably harder.

## **Conclusion**

Climate action fails not chiefly because people are unconcerned or uninformed, but because climate change resists ordinary perception. It is hard to see, localize, and connect to immediate consequences. That mismatch distorts the ethics of response. Moral attention slides toward the individual because the individual is visible and available. Meanwhile the dominant share of emissions continues to flow through industrial structures that require political intervention, not just personal sincerity. There is an optimistic outlook here. If the problem is structural, it is at least identifiable. The repeated shortfall of climate action need not be explained by a total absence of moral concern. It can be explained by a bad fit between human perceptual habits, inherited moral frameworks, and the architecture of contemporary governance. That may be discouraging, but it is traceable. Institutions can be redesigned, political narratives can be altered, leadership can perform the legibility function that this crisis requires. There is also a more pessimistic outlook. Curtin is right to describe climate change as a wicked problem, one whose scale, delay, and nonlinearity may exceed the capacities of the institutions now available to manage it, as well as our own.[22] The mismatch may not be fully solvable, political systems may remain too short-term, carbon economies too entrenched, and human beings too prone to discount the future when the present still feels livable. That tension should not be glossed over and belongs at the end of serious climate ethics; multiple solutions are the path forward here, not a single answer. What neither outlook changes is the ethical obligation. We are responsible to people who are not here to bargain with us. Future generations will inherit altered coastlines, food systems, disease burdens, and possibilities for ordinary life. They will live with consequences they did not choose and cannot retroactively refuse. That means responsibility cannot be confined to direct causation, personal intention, or the scale of individual virtue. We

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[21] Northcott, *Political Theology*, 202–3.
[22] Curtin, "Lotus among the Flames," 22.

remain obligated even when the causal chain is diffuse and the probability of success is uncertain. Practical substitutes to fossil fuel still matter, and no serious transition happens without them. But substitutes alone do not solve the central moral and political failure diagnosed here. Unless leaders make the crisis legible and redirect action toward the structures producing substantial emissions, the future will continue to absorb costs that the present was unwilling to name.

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